

Online Algebra, Logic and Topology Seminar

(CMUC, University of Coimbra)

Speaker: Giuseppe Metere (DeFENS, Università degli Studi di Milano, Italy)

Date: October 20th, 2026

Time: 15:00 (Europe/Lisbon)

Place: Google Meet

Understanding the Mundici functor (my attempt!).

Abstract. Mundici's equivalence between MV-algebras and lattice-ordered abelian groups with strong unit is usually presented through the construction of the monoid of good sequences. While effective, this construction can appear rather ad hoc from a categorical point of view.

In this talk, we reinterpret the first step of Mundici's construction through a universal property. This yields a conceptual characterisation of the monoid of good sequences and places the construction in a more formal setting.

We then explain how this universal property leads naturally to a profunctorial perspective on the Mundici functor, and we discuss how the combinatorics of good sequences can be understood through a normalization process reminiscent of polynomial multiplication.

This is joint work with Sara Lapenta and Luca Spada.